Quotient Rule

Exam code: 9MA0

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Updated on 20 June 2025

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Quotient rule

What is the quotient rule?

- The quotient rule is a formula that allows you to differentiate a **quotient of two functions** (ie one function divided by another)
- If $y = \frac{u}{v}$ where u and v are functions of x then the quotient rule is:

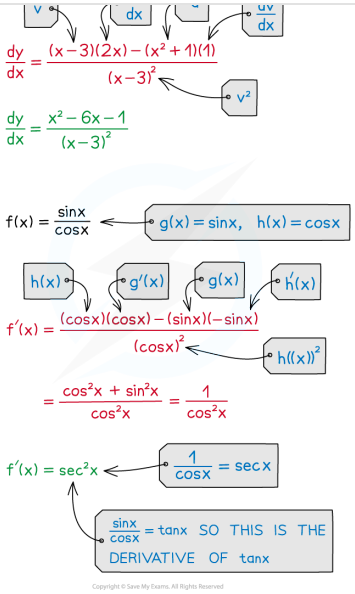
$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

- In **function notation**, if $f(x) = \frac{g(x)}{h(x)}$ then the quotient rule can be written as:

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Examiner Tips and Tricks

- The quotient rule formula is in the formulae booklet – you don't have to memorise it.
- Be careful using the formula – because of the minus sign in the numerator the order of the functions is important.
- Look out for functions of the form $\mathbf{f}(x) = \mathbf{g}(x)(\mathbf{h}(x))^{-1}$
  - You could differentiate that using a combination of the **chain rule** and the **product rule** (and it can be good practice for you to try it!)
  - But it can also be seen as just a quotient rule question in disguise
    - $\mathbf{g}(x)(\mathbf{h}(x))^{-1} = \dfrac{\mathbf{g}(x)}{\mathbf{h}(x)}$

Worked Example

with respect to x.

$$f(x) = \frac{\cos 2x}{x^2 + 2x + 1}$$

REWRITE AS $f(x) = \frac{g(x)}{h(x)}$

$g(x) = \cos 2x$ $\quad$ $h(x) = x^2 + 2x + 1$

$g'(x) = -2\sin 2x$ $\quad$ $h'(x) = 2x + 2$

CHAIN RULE – AND REMEMBER MINUS SIGN WHEN DIFFERENTIATING cos

$$f'(x) = \frac{(x^2 + 2x + 1)(-2\sin 2x) - (\cos 2x)(2x + 2)}{(x^2 + 2x + 1)^2}$$

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$$

$$f'(x) = \frac{-2\sin 2x(x^2 + 2x + 1) - 2\cos 2x(x + 1)}{(x^2 + 2x + 1)^2}$$

$$= \frac{-2\sin 2x(x + 1)^2 - 2\cos 2x(x + 1)}{(x + 1)^4}$$

$x^2 + 2x + 1 = (x + 1)^2$

$$f'(x) = \frac{-2(\cos 2x + (x + 1)\sin 2x)}{(x + 1)^3}$$

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